

Phase Three: Requirements Analysis and Modelling

Tasks 3.1, 3.2, 3.3: Actors and Use case (Diagram) Analysis & Modelling

Task 3.4: Activity Model/Diagram Analysis & Modelling

Deadline: Monday 1 June 2026

Dear Students,

After the completion of the *Requirement Engineering-Elicitation/Discovery phase*, each group is required to undertake tasks of Phase three: *Requirement Analysis & Modelling*. In this phase, each group is required to undertake, complete and submit the above tasks as described below.

Task 3.1- [Group Task] Actor Analysis:

Identify all ACTORS. Write Actors and their semantic descriptions (Examples on slides 24+26 [in L4-UML]).

Task 3.2- [Group Task] Use-Case Modelling:

Identify all USE-CASEs, and then Draw an overall/main USE-CASE diagram.

Task 3.3- [Individual Task] Use-Case Specifications:

Identify/choose **four or five** key USE CASES and write a DETAILED description of each chosen use case. **Each member** of the group should write a **(detailed) use case description** of **One** use-case. [You can find examples on slides 64, 65 in L4-UML slides]

Task 3.4- [Group Task] Activity Modelling:

Draw **overall/main** ACTIVITY diagram that shows the overall system use-cases and how they interact to achieve main business services, using swimlanes format.

=> For Group tasks, the group must discuss and then decide who would lead the task and the other members that should contribute to the task, e.g., in reviewing, discussing, re-drawing, finalising etc. Name of the Leader (of the group task) and contributing members should be clearly written on [the top of] the task, e.g., Lead: [name]; contributors: [name: reviewing], [name: discussion] etc. Tasks that have no names written clearly on them, will be disadvantaged.

=> For Individual tasks, the group must discuss and then allocate the individual task [e.g., use case specifications] to each member to complete fully- through the project manager. The name of the group member, who has done the task, should be written on [the top of] the task. Tasks that have no names written clearly on them, will be disadvantaged/ignored.

=>Diagrams must be drawn using a UML authoring tool. The group is welcome to use any UML tool of preference, that supports UML 2.x notations, (see below on UML authoring

tools), but make sure all members of the group use the **same UML tool** and the same tool is used throughout the project- members that do not use the same tool would be disadvantaged.

SUBMISSION, of final versions for all tasks, is on **the above deadline**. You are required to submit an e-copy on Ritaj- as a reply to this message, ALL tasks (3.1-3.4 above) in **ONE PDF file**, its filename as "*G [Number]-Phase3*". All UML diagrams must be drawn using (the same) UML authoring/modelling tool.

UML Authoring Tools:

- Enterprise Architecture is my favourite UML authoring tool, however, using any other UML authoring tool is acceptable.
- Two examples of UML drawing tools that offer free educational licenses are included below. However, these are essentially drawing tools and not fully UML compliant tools, i.e., they would draw UML diagrams but they will not enforce or conform to its grammar [to use, make sure to register through the below links and use your official university email]:

1- Lucid.app/lucidchart:

<https://help.lucid.co/hc/en-us/articles/360049831771-Sign-up-for-a-free-Educational-account>

2- Astah: <https://astah.net/products/free-student-license/>

- The use of AI-based UML modelling tools for **AI-generated UML diagrams** is **PROHIBITED**. AI-generated work or diagrams will receive ZERO, and may warrant a transfer to a disciplinary committee.